# Neural Selectivity in the Temporal Domain of the Auditory Cortex in Aged Rhesus Macaques

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## Introduction

- Age-related hearing deficits are one of the largest problems currently afflicting the geriatric population.
  - Age-related hearing loss, also known a presbycusis, begins in middle age and affects between 5-10% of individuals in their 50's and more than 50% of individuals 70 and older.
- These deficits are correlated with social isolation, depression, and neurodegenerative disorders such as Alzheimer's disease, Parkinson's disease and dementia.
- Those affected have difficulty discerning speech, particularly in acoustically challenging environments.
- Rhesus macaques have homologous areas in the auditory cortex and large overlap in range of hearing that allow for their use as an ideal model organism to study presbycusis in humans (fig. 1).
- Auditory cortex can be divided between rostral and caudal regions, referring to the frontmost and backmost regions respectively (fig. 2).
- 2 types of experiments are used based on 2 of the processing strategies used in auditory processing: temporal and spatial processing (fig. 3).
  - Temporal processing examines synchronicity with sound stimuli.
  - Spatial processing examines the preference of neurons to particular directions.

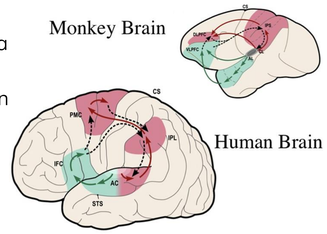



Figure 1. Homology of human and monkey brain

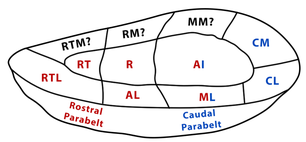



Figure 2. Rostrocaudal division of auditory cortex. Red refers to rostral and blue to caudal

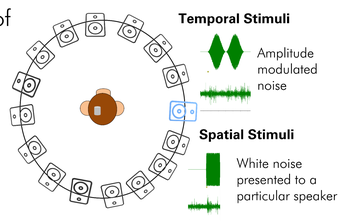



Figure 3. Experimental setup and two types of stimuli

## Hypothesis

Neural activity will be more selective in the rostral regions of the temporal domain than in the caudal regions of the temporal domain.

## Methods

Neuronal data from spatial and temporal experiments on rhesus macaques → Generate peristimulus time histograms (PSTHs) → Analyze for responsiveness, BMF, & abnormal neural responses → Generate mean spike rate and vector strength statistical data

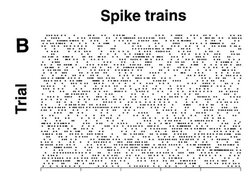



Figure 4. Spike train depicting neuronal response traces to a temporally modulated stimulus [1]

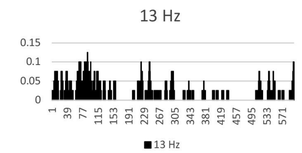



Figure 5. Typical PSTH of a temporally modulated 13 Hz tone

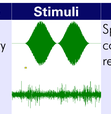
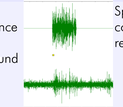
| Experiment Type | Stimuli | Presentation | Sound type |
|---|---|---|---|
| Temporal processing examines synchronicity with sound stimuli. |  | Speaker contralateral to recording cylinder | Amplitude Modulated Noise |
| Spatial processing examines a preference for particular frequencies and sound intensities. |  | Speaker contralateral to recording cylinder | Bandpass/ Broadband Noise |

Figure 7. Types of data analyzed and their corresponding presentation and sound types

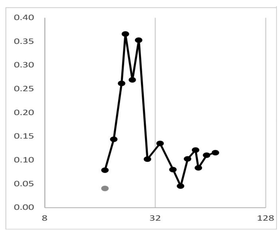



Figure 6. Typical vector strength graph used to analyze abnormal neural responses

## Results

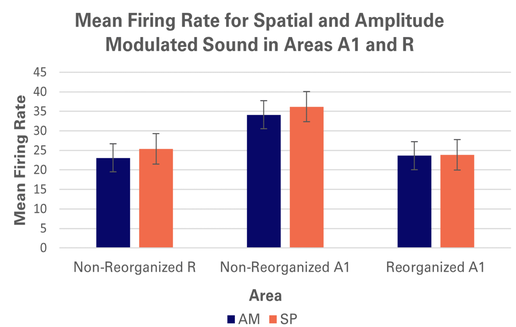



Fig 8. *Clustered column depicting mean spike rate for spatial and amplitude modulated experiments in non-reorganized areas of R and both non-reorganized and reorganized areas of A1*

## Results

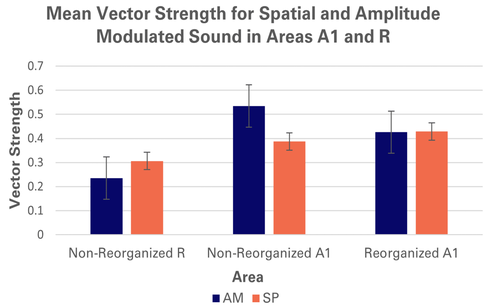



Fig 9. *Clustered column depicting vector strength for spatial and amplitude modulated experiments in non-reorganized areas of R and both non-reorganized and reorganized areas of A1*

## Conclusion

- Hypothesis is supported – higher neural selectivity in area R as compared to more caudal area A1 is indicated by lower average spike rate.
- Lower firing rate in reorganized A1 than non-reorganized A1 suggests higher neural selectivity in reorganized areas.

## Future Directions

- Data previously recorded in rostrocaudal areas AL and ML of auditory cortex will be analyzed using similar methods as detailed here.
- Neural activity in response to spectrally modulated sound will be analyzed in same areas of the auditory cortex as detailed here.

## References

1. De Ruyter van Steveninck, R. R., Lewen, G. D., Strong, S. P., Koberle, R., & Bialek, W. (1997). Reproducibility and Variability in Neural Spike Trains. *Science, 275*(5307), 1805-1808. doi:10.1126/science.275.5307.1805

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